

National University of Singapore
School of Computing
CS1101S: Programming Methodology
Semester I, 2026/2027

S2

Source Academy Burger Joint

Note

For all the problems below, you can assume that the input arguments to your functions are of the correct types and within the correct range of values.

Problems:

1. Suppose we're designing a point-of-sale and order-tracking system for a new burger joint. It is a small joint and it only sells 4 options for combos: Classic Single Combo (hamburger with one patty), Classic Double with Cheese Combo (2 patties), Classic Triple with Cheese Combo (3 patties), and Avant-Garde Quadruple with Guacamole Combo (4 patties). We shall encode these combos as 1, 2, 3, and 4 respectively. Each meal can be *biggie-sized* to acquire a larger box of fries and drink. A *biggie-sized* combo is represented by 5, 6, 7, and 8 respectively, for combos 1, 2, 3, and 4 respectively.

Write a function named `biggie_size` which when given a regular combo returns a *biggie-sized* version.

2. Write a function named `unbiggie_size` which when given a *biggie-sized* combo returns a non-*biggie-sized* version.

3. Write a function named `is_biggie_size` which when given a combo, returns `True` if the combo has been *biggie-sized* and `False` otherwise.

4. Write a function named `combo_price` which takes a combo and returns the price of the combo. Each patty costs \$1.17, and a *biggie-sized* version costs \$0.50 extra overall.

5. An order is a collection of combos. We will encode an order as each digit representing a combo. For example, the order 237 represents a Double, Triple, and *biggie-sized* Triple. Write a function named `empty_order` which takes no arguments and returns an empty order which is represented by 0.

6. Write a function named `add_to_order` which takes an order and a combo and returns a new order which contains the contents of the old order and the new combo. For example, `add_to_order(1, 2)` returns 12.

7. Write a function named `last_combo` which takes an order and returns the last combo. For example, `last_combo(321)` returns 1.

8. Write a function named `other_combos` which takes an order and returns a new order without the last combo. For example, `other_combos(321)` returns 32.